

Kevin Lu

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EDUCATION

Northeastern University
B.S. in Computer Science & Mathematics, Honors Program

Expected May 2026
GPA: 3.96/4.00

RESEARCH EXPERIENCE

Bau Lab, Interpretable Neural Networks — Northeastern University
Undergraduate Researcher

June 2024 — Present

- **Concept Erasure in Diffusion Models:** Developed probing framework for diffusion models using noisy trajectory perturbations and latent space classifier guidance. (Published at NeurIPS 2025, CVPR 2025 Visual Concepts Workshop)
- **Layer-Wise Analysis of ESM:** Identified intermediate layers as optimal for TCR-peptide binding prediction using difference of means analysis, linear probing, and component ablations. (Presented at NEMI Workshop 2025)
- **Mechanistic Interpretability in AI Protein Folding:** Performed gradient attribution and patching interventions to isolate components of ESMFold that relate to hairpin structures in proteins. (In Progress)

Takeda Pharmaceuticals
Machine Learning Research Intern

June 2024 — Present

- **Distributed Language Model Training:** Finetuned ESM-2 protein language model for masked language modeling inference on antibody sequences on 8xA100 node, improving CDR region reconstruction accuracy by 32%
- **Sparse Autoencoders for Antibody Optimization:** Trained sparse autoencoders to disentangle biologically interpretable antibody features and applied feature steering during ESM inference, generating over 10,000 high-affinity candidates; 192 variants experimentally validated with 20% increased binding specificity. (Manuscript In Preparation)

Neural Systems Group - Harvard Medical School
Undergraduate Researcher

July 2023 — May 2024

- **Signal Processing for Biomedical ML:** Implemented Kalman filtering algorithms to extract features from ECG and PPG signals and trained decision trees for blood pressure prediction (Presented at Northeastern RISE 2024 Expo).

PUBLICATIONS

Published Papers

- Kevin Lu, Nicky Kriplani, Rohit Gandikota, Minh Pham, David Bau, Chinmay Hedge, & Niv Cohen (2025). *When Are Concepts Erased From Diffusion Models?* In Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS 2025).
- Kevin Lu, Nicky Kriplani, Rohit Gandikota, Minh Pham, David Bau, Chinmay Hedge, & Niv Cohen (2025) *Where Do Erased Concepts Go?* In CVPR 2025 2nd Workshop on Visual Concepts.

Manuscripts

- Kevin Lu, Shipra Malhotra. (in preparation). *Sparse Autoencoder Feature Steering Enables Targeted Antibody Optimization.*

PRESENTATIONS & CONFERENCES

NeurIPS 2025 (39th Conference on Neural Information Processing Systems)

December 2025

K. Lu, N. Kriplani, R. Gandikota, M. Pham, D. Bau, C. Hegde, N. Cohen *When Are Concepts Erased From Diffusion Models?*

New England Mechanistic Interpretability (NEMI) Workshop

August 2025

Kevin Lu, David Bau *To Bind or Not to Bind: A Layer-Wise Dissection of Binding Information in ESM*

CVPR 2025 (3rd CVPR Workshop on Generative Models for Computer Vision)

June 2025

K. Lu, N. Kriplani, R. Gandikota, M. Pham, D. Bau, C. Hegde, N. Cohen *Where Do Erased Concepts Go?*

CVPR 2025 (Second Workshop on Visual Concepts)

June 2025

K. Lu., N. Kriplani, R. Gandikota, M. Pham, D. Bau, C. Hegde, N. Cohen *Where Do Erased Concepts Go?***RISE: Research, Innovation, Scholarship, and Entrepreneurship Expo (Northeastern University)**

April 2024

Kevin Lu, Ye Yang, Quan Zhang *Predicting Blood Pressure Using AI Models for Physiological Signals***INDUSTRY EXPERIENCE**

Babel Street

July 2024 — December 2024

Machine Learning Engineer Co-op

- **High-Throughput NLP Pipeline:** Designed multithreaded data streaming service that filters, embeds, and indexes over 1 TB of multilingual news data per week into an Elasticsearch vector database for downstream analysis.
- **Agentic Annotation System:** Developed a multi-agent LLM annotation framework using the LangGraph architecture, automating data labeling workflows and reducing annotation costs by 25%.

PROJECTS

Latent Space Exploration for Concept Erasure in Diffusion Models

February 2025 — May 2025

Course Project, MATH 7223: Riemannian Optimization

- **Latent Perturbation Optimization:** Optimized perturbation vectors to recover erased concepts in diffusion models, using Taylor-series loss landscape analysis to characterize gradient plateaus
- **Geometry-Aware Concept Recovery:** Applied Riemannian pullback metrics to model local diffusion manifolds and extract semantically meaningful latent directions via Jacobian SVD, improving erased-concept recovery from 7% to 43%.

ConcussionMute

June 2024 — December 2024

- **Signal Processing for Music:** Trained a Transformer model to isolate and suppress percussive components in audio signals, reducing high-frequency drum noise for users with sound sensitivity.

LEADERSHIP & SERVICE

Reviewer, ICLR (Geometric Deep Learning)

Nov 2025

Reviewer, NeurIPS Mechanistic Interpretability Workshop (Linear Probing Papers)

Aug 2025

President, Northeastern Veritas Forum Chapter

Sep 2023 – Present

Community Lead, InterVarsity Christian Fellowship

Sep 2023 – Present

Mandarin Teaching Aide, Dr. Martin Luther King Jr. K–8 School

Jan 2023 – Apr 2023

Mandarin Elder Service Intern, Action for Boston Community Development

Oct 2022 – Jan 2023

AWARDS

Khoury Travel Award (2X) \$1000 Travel grant for undergraduate conference presentations

July 2025, October 2025

John Martinson Honors Scholarship Annual \$40,000 merit scholarship

September 2022 — May 2026

SKILLS

Programming Languages: Python, Java, C++, MATLAB, Bash, JavaScript**Tools & Frameworks:** PyTorch, Pandas, Transformers, LangChain, Docker, BioPython, Accelerate, Scikit, FastAPI**Data & Cloud:** HuggingFace, S3, RabbitMQ, ElasticSearch, DynamoDB, Amazon Web Services (AWS)